



Department of Building and Environmental Technology
Division of Energy and Building Design

Assignment 1 – Moisture Safety Design – Autumn 2024

VBFN15 – Moisture Safety Design of Buildings

Group 1

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1 Introduction

Exterior walls shall protect the interior from exterior climate. In Lund, climate can be temperate with rainfall throughout the year. Hence, indoor and outdoor temperatures differ and relative humidity levels can reach 87.93% in December and 71.07% in May, which can cause moisture problems [1]. A high load of wind-driven rain (WDR) is also seen, leading to rain penetration in the façade [2].

To choose the right materials for exterior building components, identifying risks of condensation and mould-growth at the planning stage by conducting a performance simulation regarding moisture and heat transfer helps to optimise the design. Favourable conditions for mould growth are temperatures between 0-50°C and a relative humidity of 75-80% over a certain period of time [3].

2 Methodology

A given exterior wall (referred to as “base case”) was simulated in WUFI 6.8 for three years. WUFI incorporates all important standards, such as EN 15026 for indoor climate and DIN 4108-3 for the calculation of WDR [4]. The results were analysed using the shown graphs to identify problems regarding moisture safety.

Investigations on effects on these problems by changing materials and layering were conducted by qualitative comparisons to the base case. The result is an improved wall system, including practical thoughts like soft filling insulation between the wooden studs, OSB planks for bracing, and an installation layer to protect the vapour barrier.

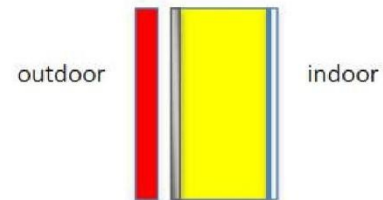
The building orientation was set to south-west to simulate the worst-case scenario regarding wind driven rain. Variations in temperature, the pattern of humidity, and global radiation were simulated using WUFI’s climate data for Lund, LTH.

3 Results

3.1 Exterior wall base case

The given wall has a total thickness of 0,465 m and a U-Value of 0,124 W/(m² K).

120 mm	Solid Brick Masonry
30 mm	Air Layer (without additional moisture capacity)
1 mm	weather resistive barrier (sd=0,1m)
300 mm	Mineral Wool (heat cond.: 0,04 W/mK) in between wooden studs
1 mm	vapour retarder (sd=100m)
13 mm	Interior Plaster (Gypsum Plaster)



3.1.1 Issues for moisture safety

1. Destruction and risk of mould growth; constantly high moisture in the brick façade at temperatures mostly between 0-20°C (Fig. 1)
2. Risk of condensation in the outermost layer of the insulation; RH > 95% long term (Fig. 2)
3. Risk of mould growth in the insulation; high RH at temperatures > 20°C long term (Fig. 3)

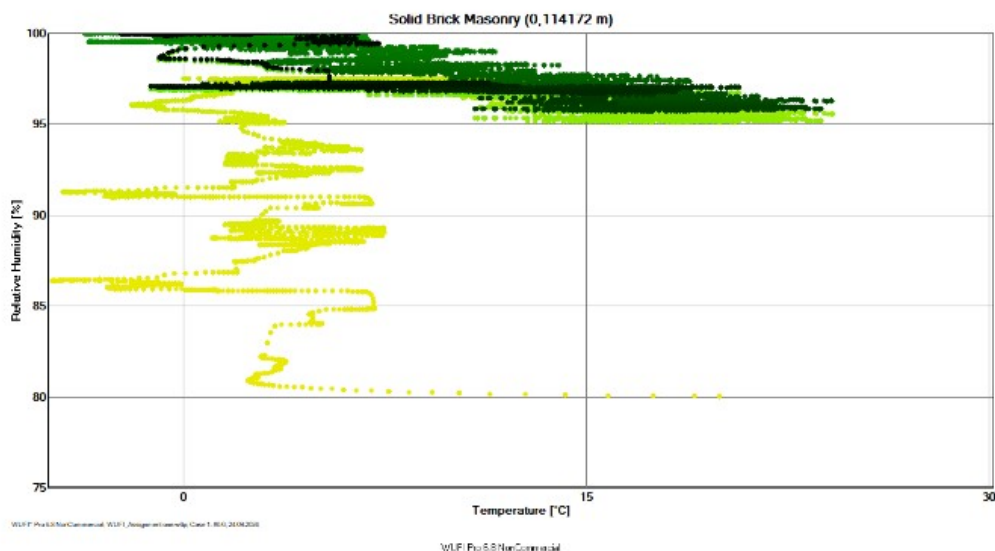


Fig. 1 RH/T on the dry side of the brick façade. The colour flow from bright to dark represents start to end of the simulation.

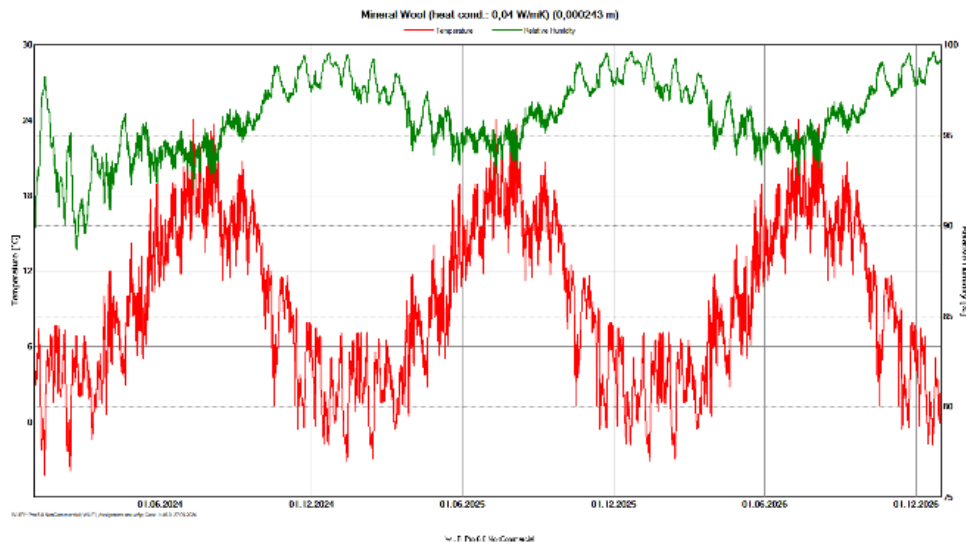


Fig. 2 RH and T over time in the outermost layer of the insulation, base case.

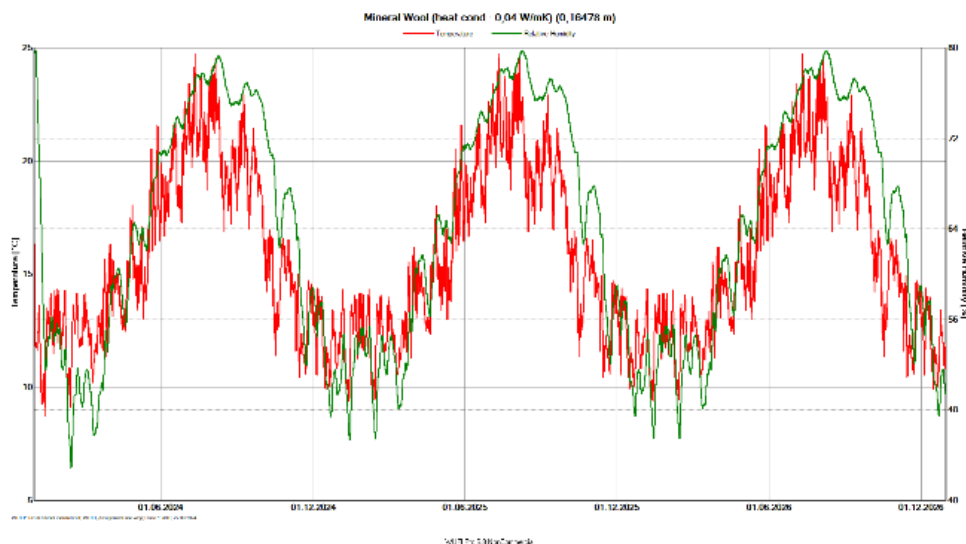


Fig. 3 RH and T over time in the middle layer of the insulation, base case.

3.2 Assessing issue 1 – Constantly high moisture in brick façade

Two main changes improved moisture control: adding a ventilation air-gap and increasing the thickness of the cavity to 40 mm. Designing the air-gap with 50 air changes per hour (common for brick facades [5]) helped a lot to efficiently remove moisture (Fig. 4). Aluminium cladding was considered for extra protection but was avoided to preserve the buildings appearance.

Other measurements that were applied (i.e. different wind barriers, insulation materials, layers, ...) did not show a significant effect on this issue, since the source of the moisture is rain, not diffusion. Still, a significant reduction of moisture in the brick was achieved in our final case (Fig. 1, Fig. 5), but a risk of mould growth is given in the end.

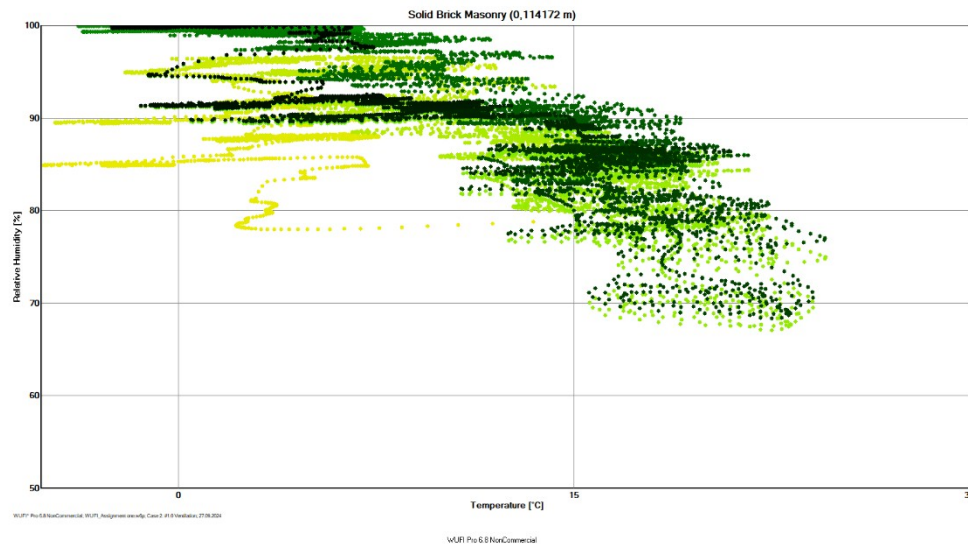


Fig. 4 RH/T on the dry side of the brick façade, base case plus ventilated air-gap.

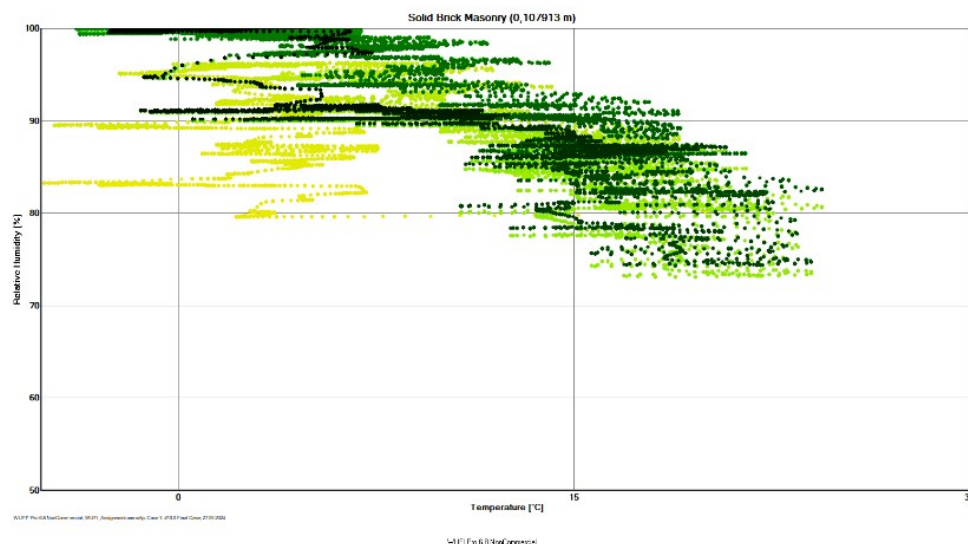


Fig. 5 RH/T on the dry side of the brick façade, final improved version.

3.3 Assessing issue 2 – Risk of condensation in outermost layer of insulation

Firstly, less moisture shall enter the insulation layer. Vapour diffusion from the interior can be buffered (clay render, installation layer, ...) and blocked (vapour barrier). Changes regarding the moisture buffering did not show a recognisable effect, neither did applying a stronger vapour barrier ($s_d=1500\text{m}$). The simulations have shown that the weather barrier is a must to block occasional moisture transfer from the exterior through the air-gap.

Secondly, measurements can aim on helping the insulation to dry out. The biggest improvement was achieved by having a ventilated air-gap. Moreover, vulnerable material was removed from sensible areas by adding less hygroscopic insulation outside the weather barrier. An air- and water-resistant weather barrier with a lower s_d -value was chosen. The main insulation material was changed to store less moisture and to have a constant heat transfer coefficient. After all, the risk of condensation in the insulation was eliminated (Fig. 6, Fig. 7).

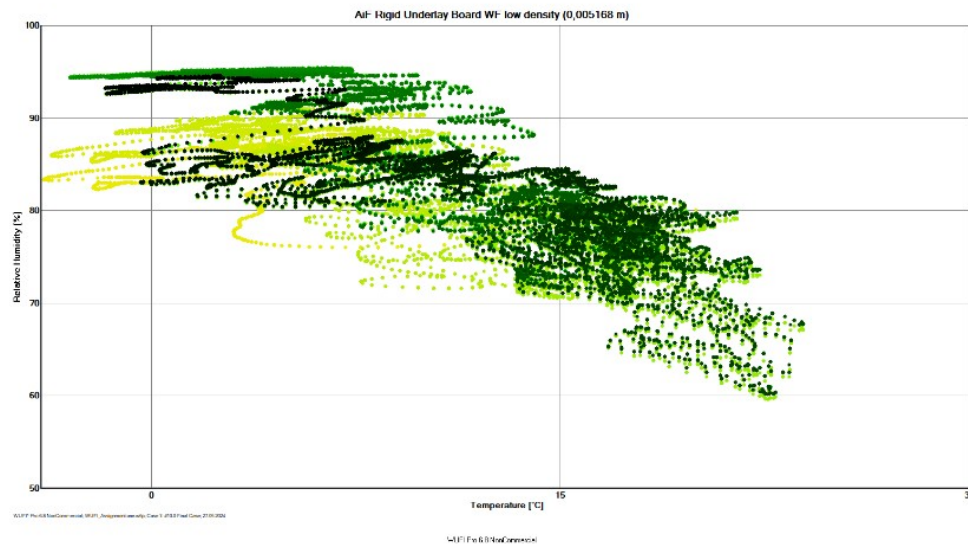


Fig. 6 RH/T in the wooden insulation board in between the ventilated air-gap and the weather barrier with decreased thickness of the main insulation.

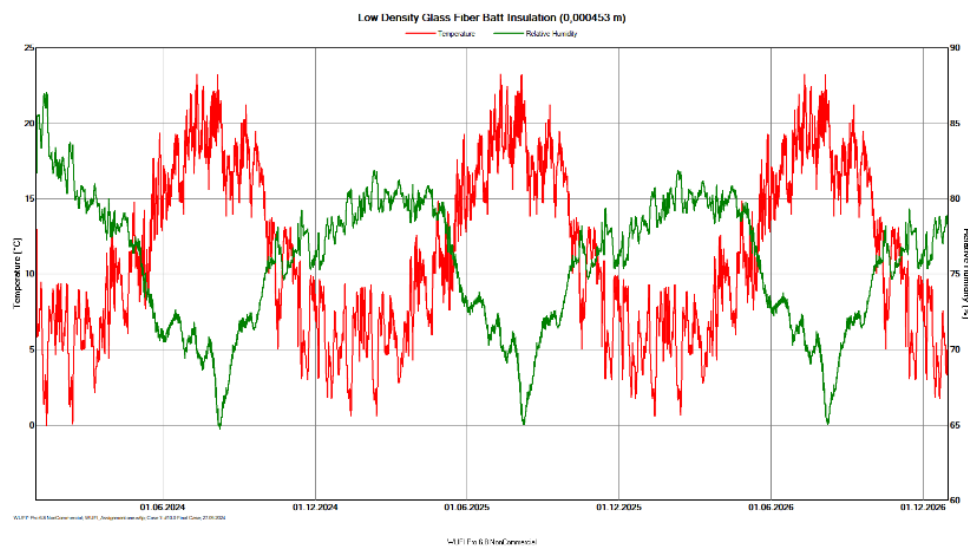


Fig. 7 RH and T over time in the outermost layer of the insulation, final improved wall system.

3.4 Assessing issue 3 – Risk of mould growth within the insulation

The improvements made to assess issue 2 are also beneficial for issue 3. Increasing the s_d -value of the vapour barrier did not show any effect as well. But when the wood-fibre board, weather barrier and thinner insulation layer were applied, the relative humidity decreased. Having ventilation in the air-gap showed the same impact. Comparing the final improved version with the base case (Fig. 3, Fig. 8), a large decrease of the relative humidity in the middle layer of the insulation to a stable level of below 65% is seen. Hence, mould growth is avoided.

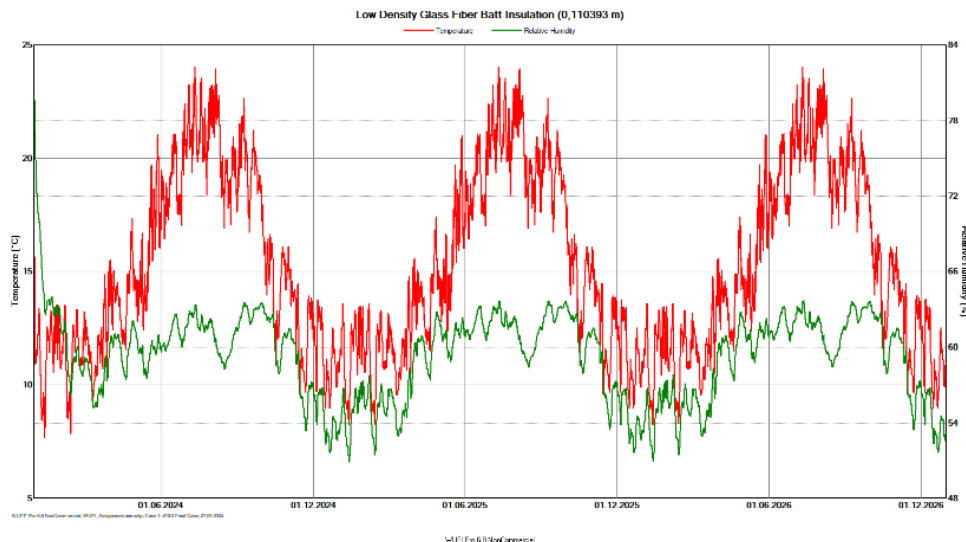
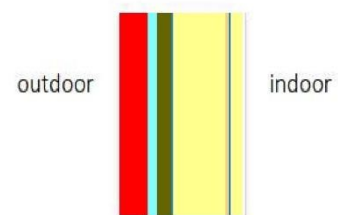


Fig. 8 RH and T over time in the middle layer of the insulation, final improved version.

3.5 Exterior wall improved version

The suggested improvements add up to the following components with a total thickness of 0,52 m and a U-Value of 0,121 W/(m² K).

120 mm	Solid Brick Masonry
40 mm	Air Layer (without additional moisture capacity) ventilation 50 Air Changes per hour
60 mm	AiF Rigid Underlay Board WF low density ($\lambda=0.043$ W/mK)
1 mm	DELTA-VENT S (sd=0,055m)
220 mm	Low Density Glass Fiber Batt Insulation ($\lambda=0,043$ W/mK) in between wooden studs
15 mm	OSB 3
1 mm	vapour retarder (sd=100m)
50 mm	Installation layer; Low Density Glass Fiber Batt Insulation ($\lambda=0,043$ W/mK)
13 mm	Gypsumboard, interior



4 Conclusion

Highly insulated exterior walls like the given one show difficulties in handling moisture to not damage the structure nor having conditions that allow mould growth. The harsh and wet climate of Lund, Sweden increases the moisture impact. By applying different measurements, a safe design is possible. Over all a ventilated air-gap appears to be most impactful, as well as a good vapour barrier on the inside and a weather resistant but breathable membrane on the outside of the insulation layer.

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